

Name
Date
Course

Evaluating Composite Trigonometric Functions (Homework)

Find the value of each expression **without a calculator**.

① $\cos(\sin^{-1} \frac{1}{2})$

② $\tan(\sec^{-1} \sqrt{2})$

$$\textcircled{3} \csc(\cot^{-1} \frac{1}{\sqrt{3}})$$

$$\textcircled{4} \csc(\sin^{-1} 1)$$

$$\textcircled{5} \cot(\cos^{-1}(-\frac{\sqrt{3}}{2}))$$

$$\textcircled{6} \csc(\sec^{-1}(-\sqrt{2}))$$

$$\textcircled{7} \sin(\csc^{-1}(-\frac{2}{\sqrt{3}}))$$

$$\textcircled{8} \sec(\tan^{-1}(-1))$$

$$\textcircled{9} \cos(\cos^{-1} \frac{\sqrt{3}}{2})$$

You should be able
to do these quickly.

$$\textcircled{10} \csc(\csc^{-1} 2)$$

$$\textcircled{11} \cot(\cot^{-1}(-\sqrt{3}))$$

Write an equivalent expression for each problem without trigonometric notation. Assume that x is positive.

$$\textcircled{12} \sin(\cos^{-1} \frac{1}{x})$$

$$\textcircled{13} \cot(\csc^{-1} x)$$

⑭ $\sec(\tan^{-1} \frac{1}{x})$

Homework Answers

① $\frac{\sqrt{3}}{2}$

② 1

③ $\frac{2\sqrt{3}}{3}$

④ 1

⑤ $-\sqrt{3}$

⑥ $\sqrt{2}$

⑦ $-\frac{\sqrt{3}}{2}$

⑧ $\sqrt{2}$

⑨ $\frac{\sqrt{3}}{2}$

⑩ 2

⑪ $-\sqrt{3}$

⑫ $\frac{\sqrt{x^2-1}}{x}$

⑬ $\sqrt{x^2-1}$

⑭ $\frac{\sqrt{x^2+1}}{x}$